

4. Logic Gates and Boolean Algebra

26 April 2023

Notes

1. Boolean values could be either 1 or 0 means either on or it could be True or False. In an electronic circuit it could corresponds to High voltage and Low voltage respectively. There are no other values in Boolean algebra like 2 or $\frac{1}{2}$.
2. Difference between Boolean algebra and normal algebra.
Normal algebra can have any value while Boolean algebra can have value 0 or 1. Boolean algebra calculation differ from the algebra as in Boolean algebra $1+1 = 1$ (High voltage + High Voltage results a high voltage).

3. Past question papers

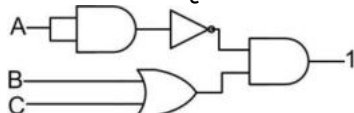
1. Select the logical operation corresponding to the following truth table:

GIT2005MCQ06

Input 1	Input 2	Output
0	0	1
0	1	0
1	0	0
1	1	0

1)OR 2)AND 3)NOR 4)NAND

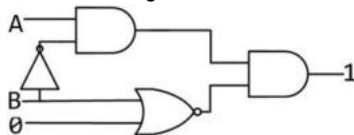
2. Find the value of the two inputs A and B such that the output is 1 and C is 0. GIT2005MCQ07



1)0 and 0 2)0 and 1 3)1 and 0 4)1 and 1

3. Find the value of the two inputs A and B such that the output is 1.

GIT2006MCQ06



1)0 and 0 2)0 and 1 3)1 and 0 4)1 and 1



In digital logic the above symbols A and B respectively represent the gates. OL2007MCQ18

1)NOR and AND 2)NOR and OR 3)OR and NOR 4)NOT and OR

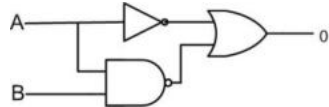
5. The following is the truth table of a certain logic gate that has two inputs A, B and an output F. The logic gates is OL2007MCQ19

A	B	F
0	0	0
0	1	0
1	0	0
1	1	1

1)OR 2)AND 3)NOR 4) NAND

6. Find the values of the two inputs A and B such that output is 0.

GIT2007MCQ07



- 1) 0 and 0 2) 0 and 1 3) 1 and 0 4) 1 and 1

7. Which Boolean relation below gets verified by the truth table given in Figure OL2008MCQ15

A	$\bar{A}$	Output
1	0	1
0	1	1

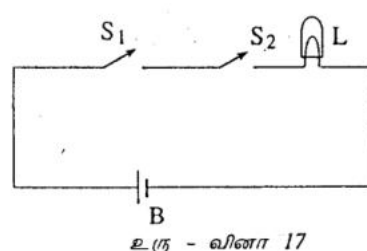
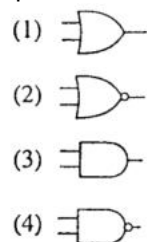
- 1) $A + \bar{A} = 1$ 2) $A \cdot \bar{A} = 0$ 3) Both 1 and 2 4) $A + A = 1$

8. 

Which one of the following circuits indicates the construction of a NAND logic gate by using the above logic gates? OL2008MCQ16



9. Which of the following digital logic gates can be used to characterize the operation of the electrical circuit shown in Figure? OL2008MCQ17



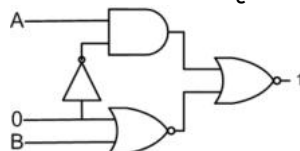
L - விளக்கு
S₁ - ஆனி 1
S₂ - ஆனி 2
B - பற்றறி (மின்கலம்)

உரு - வினா 17

10. $A + (\bar{A} \cdot B)$ is equivalent to. GIT2008MCQ05

- 1) A 2) B 3) A+B 4) A.B

11. Find the value of the two inputs A and B respectively such that the output is 1. GIT2008MCQ07.



- 1) 0 and 0 2) 0 and 1 3) 1 and 0 4) 1 and 1

12. Which one of the following Boolean relations is verified by the given truth table? OL2009MCQ15

A	1	Output
0	1	1
1	1	1

- 1) $A + A = A$ 2) $A \cdot A = A$ 3) $A + A = A$ and $A \cdot A = A$ 4) $A + 1 = 1$

13. "Which of the digital logic symbols given in the table represents NOT logic? OL2009MCQ16



- 1) A and B 2) B and C 3) A and C 4) A, B and C

14. A father presents an electronic talking parrot to his son on his birthday. It can be controlled remotely by two push button switches. When both buttons are not pressed at all or pressed together, the parrot does not talk, but talks when either one of them is pressed. The digital logic that corresponds to the operation of the talking parrot is. OL2009MCQ17
1)OR 2)AND 3)NOR 4)XOR

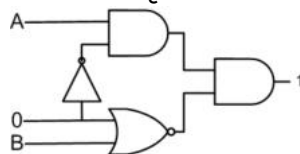
15. The logical expression $A.(A+B)$ is equivalent to GIT 2009MCQ05
1) A 2)B 3) $A+B$ 4) $A.B$

16. Select the logical operation corresponding to the following truth table. GIT2009MCQ05

Input1	Input2	Output
0	0	1
0	1	0
1	0	0
1	1	0

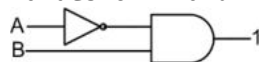
1)NOR 2)NAND 3)XOR 4)XNOR

17. Find the value of the two inputs A and B such that the output is 1. GIT2009MCQ07



1)0 and 0 2)0 and 1 3)1 and 0 4)1 and 1

18. If the output of the logic circuit in Figure 2 is 1, what should be the values of A and B, respectively? OL2010MCQ11

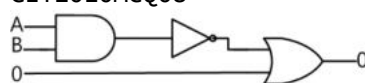


1)0 and 0 2)1 and 0 3)0 and 1 4)1 and 1

19. A student could select either string hoppers or bread to eat with dhal curry for his breakfast. Which of the following logical operations currently represents this scenario? OL2010MCQ12

1)AND followed by OR 2)OR followed by AND 3)OR followed by OR 4)AND followed by AND

20. Find the values of A and B respectively to produce 0 as the output. GIT2010MCQ08



1)0 and 1 2)1 and 1 3)1 and 0 4)0 and 0

21. Select the logical operation corresponding to the given truth table. GIT2010MCQ09

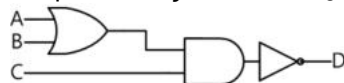
Input X	Input Y	Output
0	0	0
0	1	1
1	0	0
1	1	1

1) $(X \text{ OR } Y) \text{ AND } Y$ 2) $(X \text{ AND } Y)$ 3) $\text{NOT}(X \text{ OR } Y)$ 4) $(X \text{ AND } Y) \text{ OR } X$

22. A merchant sells a pack of three items for Rs 75. The pack contains a cup (A) and a saucer (B). The customer can select either a plate (C) or a dish (D) as the third term. Which of the following Boolean expression represents the above scenario? OL2011MCQ11

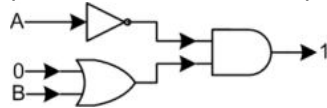
- 1) $(A \text{ AND } B) \text{ AND } C \text{ AND } D$ 2) $A \text{ AND } B \text{ OR } C \text{ OR } D$ 3) $A \text{ AND } B \text{ AND } (C \text{ OR } D)$ 4) $A \text{ OR } B \text{ OR } (C \text{ AND } D)$

23. If the final output of the given logic circuit D is 1, which of the following represents the corresponding values for the input A, B and C respectively? OL2011MCQ12



- 1) $A=1, B=1, C=1$
 2) $A=1, B=0, C=1$
 3) $A=0, B=1, C=0$
 4) $A=0, B=1, C=1$

24. What should be the input A and B respectively in the logic circuit to produce 1 as the output? GIT2011MCQ06



- 1) 0 and 1 2) 0 and 0 3) 1 and 0 4) 1 and 1

25. Consider the following truth table output of the truth table produced by GIT2011MCQ08

Input X	Input Y	Output
0	0	1
0	1	1
1	0	1
1	1	0

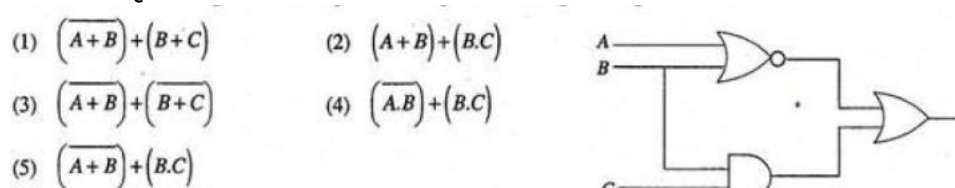
- 1) $X \text{ AND } Y$ 2) $\text{NOT}(X \text{ AND } Y)$ 3) $X \text{ OR } Y$ 4) $\text{NOT}(X \text{ OR } Y)$

26. Which of the following would be the result if the Boolean Expression is simplified by using De Morgan's Law? AL2011MCQ27

$$F(x, y) = \overline{(x + y)} \cdot (\bar{x} + \bar{y})$$

1) x 2) y 3) 0 4) 1 5) $x \cdot y$

27. Which of the following Boolean expression represents the given logic circuit? AL2011MCQ28

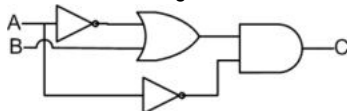


- (1) $\overline{(A+B)} + (B+C)$ (2) $(A+B) + (B.C)$
 (3) $\overline{(A+B)} + (B+C)$ (4) $\overline{(A.B)} + (B.C)$
 (5) $\overline{(A+B)} + (B.C)$

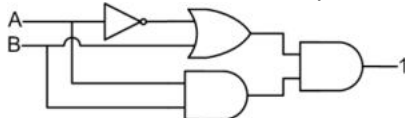
28. A question paper consists of three questions x, y and z. Candidates are required to answer two questions. Question x is compulsory and the candidate has to select one from question y and z. Which of the following represents the selection of questions by a candidate? OL2012MCQ10

- 1) $x \text{ OR } (y \text{ OR } z)$ 2) $x \text{ AND } (y \text{ AND } z)$ 3) $x \text{ AND } (y \text{ OR } z)$ 4) $x \text{ OR } (y \text{ AND } z)$

29. Consider the given logic circuit with A and B as inputs and C as an output. Which of the following is correct regarding the inputs when the output C = 1? OL2012MCQ12



- 1) A should always be 0. 2) A should always be 1
3) B should always be 0. 4) B should always be 1.
30. The out of NOT(NOT(A OR B)) is equivalent to. GIT2012MCQ03
1) NOT (A OR B) 2) A OR B 3) A AND B 4) NOT(A AND B)
31. The output of the logic circuit shown below is 1. GIT2012MCQ06
What should be the inputs of A and B respectively.



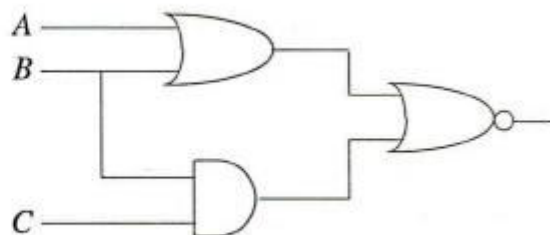
- 1) 0, 0 2) 0, 1 3) 1, 0 4) 1, 1
32. Consider the following truth table. Which of the following is represented by the symbol GIT2012MCQ08

A	B	$A \nabla B$
0	0	1
0	1	0
1	0	0
1	1	0

- Which of the following is represented by the symbol ∇ ?
1) OR 2) AND 3) NOT(OR) 4) NOT(AND)
33. What would be the result if the following Boolean expression is simplified? AL2012MCQ26
 $F(x,y) = \overline{x}\overline{y}(\overline{x}+y)(y+\overline{y})$
1) $\overline{x}$ 2) $\overline{y}$ 3) x 4) y 5) xy

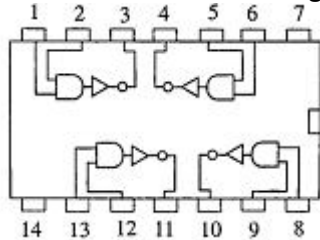
34. Which of the following expression represents the output of the given logic circuit? AL2012MCQ27

- (1) $(\overline{A+B}) + (\overline{B \cdot C})$
(2) $(\overline{A+B}) \cdot (\overline{B \cdot C})$
(3) $\overline{(A+B) + (B \cdot C)}$
(4) $(\overline{A \cdot B}) + (\overline{B \cdot C})$
(5) $\overline{(A \cdot B) + (B \cdot C)}$



35. Consider the following logical expression P, Q and R: OL2013MCQ29
P: (A > B) OR (C > D)
Q: (A < B) AND (C > D)
R: NOT (A < B)
If above A, B, C and D contain values 50, 40, 30 and 20 respectively, which of the following represents the correct outcomes of P, Q and R respectively?
1) True, False, False 2) True, False, True
3) True, True, False 4) True, True, True

36. Consider the following integrated circuit (IC) chip: 0L2013MCQ30



Which of the following is correct regarding pins 1, 2 and 3 of the above chip?

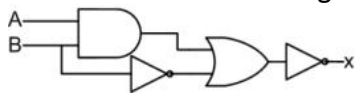
- 1) If pin 1 = 0 and pin 2 = 0 then pin 3 will be 0.
- 2) If pin 1 = 0 and pin 2 = 1 then pin 3 will be 0.
- 3) If pin 1 = 1 and pin 2 = 0 then pin 3 will be 1.
- 4) If pin 1 = 1 and pin 2 = 1 then pin 3 will be 1.

37. A truth table and the corresponding logic circuit are given below. GIT2013MCQ03.

Input A	Input B	Output
0	0	0
0	1	1
1	0	1
1	1	1



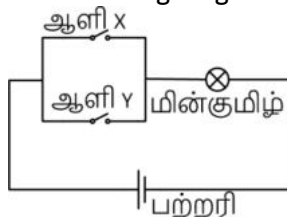
38. Consider the following logic circuit are given below. GIT2013MCQ04



If the value of the output X of the above logic circuit is 1 determine the values for the inputs A and B.

- 1) A = 0 and B = 0
- 2) A = 1 and B = 0
- 3) A = 0 and B = 1
- 4) A = 1 and B = 1

39. Consider the following circuit, which is similar to the operation of a certain logic gate.



Which of the following statement(s) is/are correct regarding the above circuit? GIT2013MCQ11

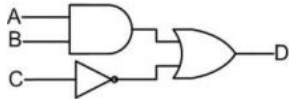
- A - The bulb illuminates when the Switch X is turned ON.
- B - The bulb illuminates when both the switch X and Y are turned ON.
- C - The bulb illuminates when the switch X is turned OFF and the switch Y is turned ON.

- 1) A and B only
- 2) B and C only
- 3) A and C only
- 4) All A, B and C

40. The boolean expression $(X+y).(x+z)$ simplifies to AL2013MCQ01

- 1) x
- 2) $x.(y+z)$
- 3) $x.y.z$
- 4) $x+y.z$
- 5) $x+y+z$

41. Consider the given logic circuit: OL2014MCQ23



If $A = 1$ and $D = 0$ then which of the following is correct?

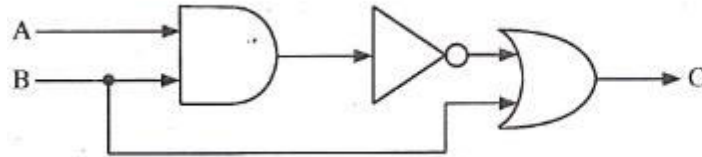
- 1) $B = 0$ and $C = 0$ 2) $B = 0$ and $C = 1$ 3) $B = 1$ and $C = 1$ 4) $B = 1$ and $C = 0$

42. Consider the following Boolean expression $D = A.B + C'$

Which of the following is correct? OL2014MCQ24

- 1) if $A = 0$ and $B = 1$ then $D = 1$ irrespective of what C is
 2) if $A = 1$, then $D = 1$ irrespective of what B and C are
 3) if $B = 0$, then $D = 0$ irrespective of what A and C are
 4) if $C = 0$, then $D = 1$ irrespective of what A and B are

43. The inputs for the following logic circuit are as shown in the following truth table: GIT2014MCQ04

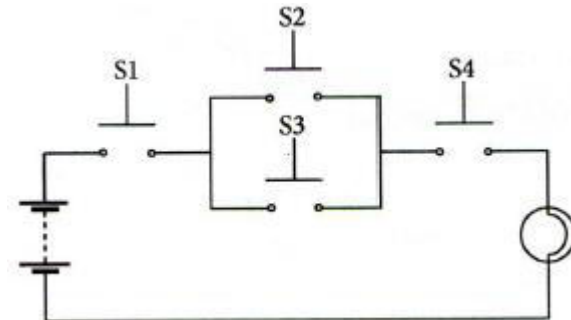


A	B	C
0	0	
0	1	
1	0	
1	1	

In the above table, what are the output values for column C from top to bottom respectively?

- 1) 0, 0, 0, 0 2) 0, 1, 0, 1 3) 1, 0, 1, 0 4) 1, 1, 1, 1

44. Consider the following circuit with four push button switches namely S_1 , S_2 , S_3 and S_4 . These four switches can either be in pushed or released states which are represented by 1 and 0 respectively. (Note: In the circuit given below, all the switches are in released state value 0). AL2014MCQ13



Which of the following Boolean expression represents the function of the bulb, if the on state of the bulb is represented by the value 1?

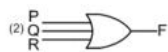
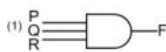
- 1) $S_1 + (S_2.S_3) + S_4$ 2) $(S_1 + S_2).(S_3 + S_4)$ 3) $(S_1.S_2) + (S_3.S_4)$ 4) $S_1.S_4.(S_2 + S_3)$
 5) $S_2 + (S_1.S_4) + S_3$

45. Consider a logic circuit with three inputs P , Q , R and one output F . The following is observed in this circuit: OL2015MCQ03

If any single input is 1, then output F is 0, irrespective of what the remaining inputs are

If all inputs are 0, then the output F is 1

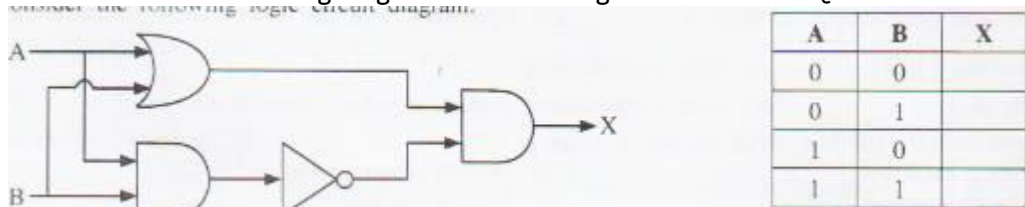
Which of the following is the most likely logic gate configuration for the circuit?



46. To qualify in an examination, candidates have to pass one compulsory subjects S_1 and one of the three optional subjects S_2 , S_3 and S_4 . Which of the following Boolean expression represents this scenario? OL2015MCQ04

- 1) S_1 AND (S_2 AND S_3 AND S_4) 2) S_1 AND (S_2 OR S_3 OR S_4)
 3) S_1 OR (S_2 AND S_3 AND S_4) 4) S_1 OR (S_2 OR S_3 OR S_4)

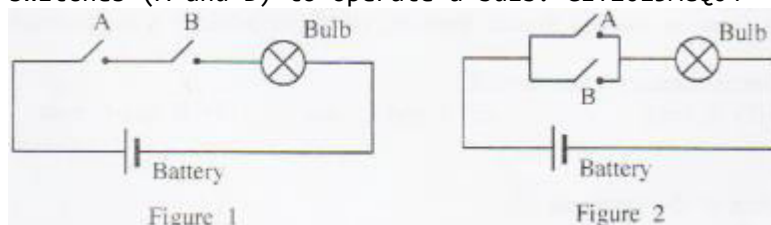
47. Consider the following logic circuit diagram. GIT2015MCQ02



The output X for the inputs given in the truth table for the above logic circuit are respectively.

- 1) 0, 0, 0, 0 2) 0, 1, 0, 1 3) 0, 1, 1, 0 4) 1, 0, 1, 0

48. Figure 1 and 2 below illustrates two circuit that use a battery and two switches (A and B) to operate a bulb. GIT2015MCQ04

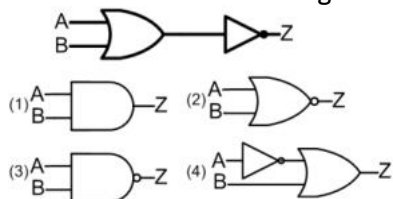


Which of the following logic gate combinations represent the operation of the above two circuits in the figures 1 and 2 respectively?

- (1) . (2) . (3) . (4) .

49. Consider the following logic circuit diagram

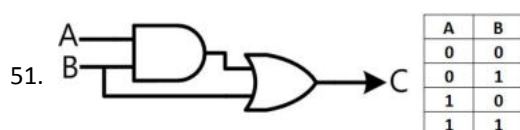
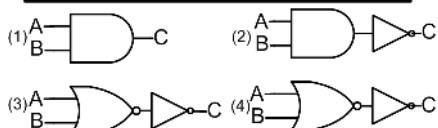
Which of the following is equivalent to the above logic circuit? OL2016MCQ06



50. Which of the following logic circuits represents the given truth table?

OL2016MCQ07

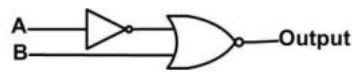
A	B	C
0	0	1
0	1	1
1	0	1
1	1	0



What are the respective outputs of the above logic circuit when the inputs given in the table are applied? GIT2016MCQ36

- 1) 0, 0, 0, 0 2) 0, 0, 0, 1 3) 0, 1, 0, 1 4) 1, 0, 0, 0

52. Which of the following truth table is equivalent to the logic circuit given below? OL2017MCQ07



A	B	Output
0	0	1
0	1	1
1	0	1
1	1	0

(1)

A	B	Output
0	0	0
0	1	0
1	0	1
1	1	0

(2)

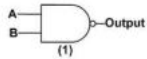
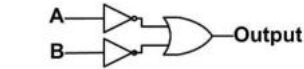
A	B	Output
0	0	0
0	1	1
1	0	0
1	1	0

(3)

A	B	Output
0	0	0
0	1	1
1	0	1
1	1	0

(4)

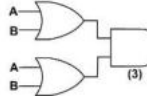
53. Which of the following logic circuits has a truth table equivalent to the truth table of the logic circuit shown on the right hand side? OL2017MCQ08



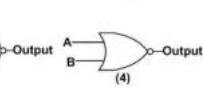
(1)



(2)

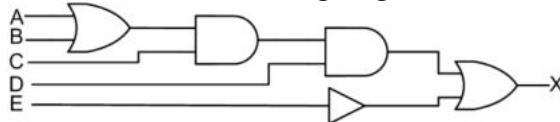


(3)



(4)

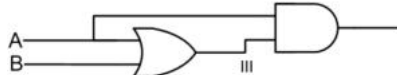
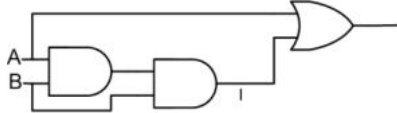
54. Consider the following logic circuit OL2018MCQ06



Which of the following is equivalent to the above circuit?

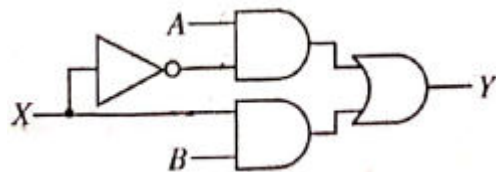
- 1) $X = (A+B).C.(D+\bar{E})$ 2) $X = (A+B).C.D+\bar{E}$
 3) $X = (A.B)+C+D.\bar{E}$ 4) $X = (A+B).(C+D)+\bar{E}$

55. For given inputs, which of the following logic circuits provide the same output? OL2018MCQ07



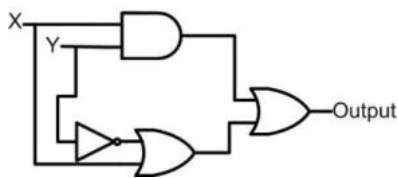
- 1) I and II only 2) I and III only 3) II and III only 4) all I, II, III

56. If 0 and 1 respectively are given as inputs for X in the following logic circuit, what would be the two respective output at Y? OL2019MCQ40



- 1) A, $\bar{B}$ 2) A, B 3) B, $\bar{A}$ 4) B, A

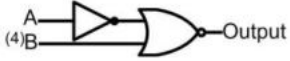
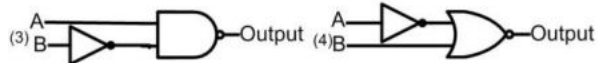
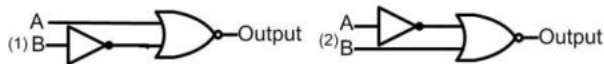
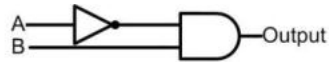
57. Which of the following Boolean expression is equivalent to the output of the given logic circuit? OL2020MCQ34



- 1) $(x \cdot y) + (\bar{y} + x)$ 2) $(x + y) \cdot (\bar{y} \cdot x)$ 3) $(x + y) \cdot (x \cdot \bar{y})$ 4) $(x \cdot y) + (y + \bar{x})$

58. Consider the following logic circuit:OL2020MCQ35

Which of the following circuits has a truth table equivalent to the above circuit?



59. Consider the following logic gate



When $B = 1$, then what would definitely be the output at Q?

- 1) A 2) $\bar{A}$ 3) $A + B$ 4) $\bar{A} + B$

60. Given below is the truth table for a logic gate with inputs A, B and Q:

A	B	Q
0	0	1
0	1	1
1	0	1
1	1	0

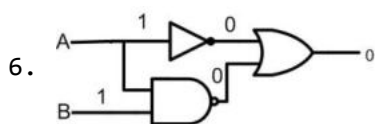
What is the logic indicated by the above truth table?

- 1) AND 2) NAND 3) NOR 4) OR

61.

Answers

- NOR
- 0 and 1
- 1 and 0
- NOT and OR
- AND



7. $A + \bar{A} = 1$

8. **NOT AND** = NAND (3rd answer)

9. AND (3rd answer)

10. $A + B$ (OL, GIT students can come to this answer from the truth table. As it refers to an OR table so it is $A + B$).

A	B	$\bar{A}$	$\bar{A} \cdot B$	$A + \bar{A} \cdot B$
0	0	1	0	0
0	1	1	1	1

1	0	0	0	1
1	1	0	0	1

AL ICT students can solve this using Boolean algebra as well.

$$A + \bar{A}.B$$

$$= A + A + \bar{A}.B$$

$$= A + A.1 + \bar{A}.B$$

$$= A + A(B + \bar{B}) + \bar{A}.B$$

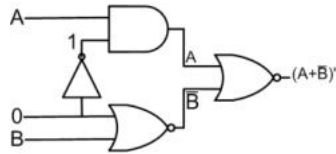
$$= A + A.B + \bar{A}.B$$

$$= A + A.B + B$$

$$= A(1 + B) + B$$

$$= A + B$$

11. 0 and 1



12. $A+1=1$

13. A and C

14. XOR

15. AL ICT students can come to the answer from the Boolean Algebra

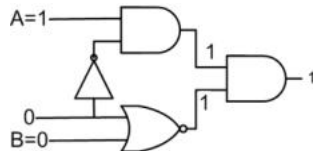
$$A(A+B) = A.A + A.B = A + A.B = A(1+B) = A$$

OL ICT students can come to this answer from the truth table.

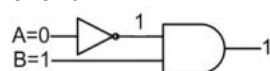
A	B	A+B	A.(A+B)
0	0	0	0
0	1	1	0
1	0	1	1
1	1	1	1

16. NOR (NOT + OR)

17. 1 and 0



18. 0 and 1



19. OR followed by AND (Tamil medium questions looks different).

20. 1 and 1

21. $(X \text{ OR } Y) \text{ AND } Y$

22. A AND B AND (C OR D)

23. $A=0, B=1, C=0$

24. 0 and 1

25. $\text{NOT}(X \text{ AND } Y)$

26. 0 (OL, GIT students draw truth table to verify)

X	Y	$\bar{X}$	$\bar{Y}$	$X+Y$	$(X+Y)'$	$\bar{X}+\bar{Y}$	$(\bar{X}+\bar{Y})'$	$(X+Y)' . (\bar{X}+\bar{Y})'$
0	0	1	1	0	1	1	0	0
0	1	1	0	1	0	1	0	0
1	0	0	1	1	0	1	0	0
1	1	0	0	1	0	0	1	0

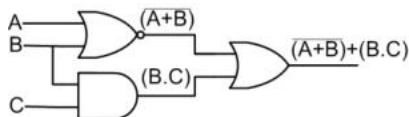
AL ICT students

$$(X+Y)' . (\bar{X}+\bar{Y})'$$

$$= (\bar{X} \cdot \bar{Y}) \cdot (X \cdot Y) \text{ (De Morgan's Law } X'' = X \text{ and } y'' = y)$$

$$= 0 \text{ (} X\bar{X} = 0 \text{ and } Y\bar{Y} = 0)$$

27. 5th answer



28. x AND (y OR z)

29. A should always be 0.

30. A OR B

31. 1, 1

32. NOT(OR)

33. OL and GIT students solve this by the truth table. AL ICT students may solve this by the following boolean algebra.

$$\begin{aligned}
 & (\bar{x} + \bar{y}) \cdot (\bar{x} + y) \cdot (y + \bar{y}) \\
 &= (\bar{x} + \bar{y}) \cdot (\bar{x} + y) \text{ (since } (y + \bar{y}) = 1) \\
 &= \bar{x} \cdot \bar{x} + \bar{x} \cdot y + \bar{x} \cdot \bar{y} + y \cdot \bar{y} \\
 &= \bar{x} + \bar{x}(y + \bar{y}) \text{ (since } y \cdot \bar{y} = 0 \text{ and } x \cdot x = x) \\
 &= \bar{x} + \bar{x} \text{ (since } (y + \bar{y}) = 1) \\
 &= \bar{x} \text{ (since } A + A = A)
 \end{aligned}$$

34. 2nd and 3rd answer.

35. True, False, True

36. If pin 1 = 1 and pin 2 = 0 then pin 3 will be 1.

37. OR Gate (3rd answer).

38. A = 0 and B = 1

39. All A, B and C

40. x+ y.z (From the truth table)

X	Y	Z	X+Y	X+Z	(X+Y).(X+Z)	X.(Y+Z)	Y.Z	X+Y.Z
0	0	0	0	0	0	0	0	0
0	0	1	0	1	0	0	0	0
0	1	0	1	0	0	0	0	0
0	1	1	1	1	1	0	1	1
1	0	0	1	1	1	0	0	1
1	0	1	1	1	1	1	0	1
1	1	0	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1

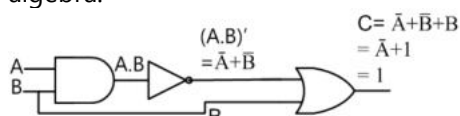
AL ICT students can solve this by using Boolean algebra

$$\begin{aligned}
 & (x+y).(x.z) \\
 &= x.x + x.z + y.x + y.z \\
 &= x + x.z + x.y + y.z \text{ (} x.x = x, y.x = x.y) \\
 &= x(1+z+y) + y.z \\
 &= x + y.z \text{ (} 1+A = 1)
 \end{aligned}$$

41. B = 0 and C = 1

42. if C = 0, then D = 1 irrespective of what A and B are.

43. 1, 1, 1, 1 (OL, GIT students can substitute the value. AL ICT students can solve with the Boolean algebra.



44. S1.S4.(S2+S3)

45. Given logic refers to NOR gate so we have to choose OR + NOT

46. S1 AND (S2 OR S3 OR S4)

47. 3rd answer (OL, GIT students can come this answer from the truth table and AL ICT can solve this by using Boolean algebra).

48. 2nd answer.

49. NOT + OR = NOR 2nd answer

50. Given truth table refers to NAND gate so 2nd answer

51. 0, 1, 0, 1

52. 2nd answer $(A'+B)' = A' \cdot B' = A \cdot B'$ or solve it with the truth table.

53. Given Boolean expression is $\overline{A+B}$.

First answer's Boolean expression is $(A+B)' = \overline{A+B}$ (as per De Morgan's law) so it's the correct answer.

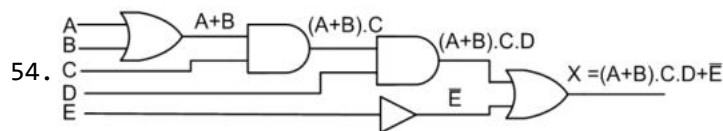
Second answer's Boolean expression is $(A+B)' + B = \overline{A+B} + B$ $((A+B)' = \overline{A+B}$ as per De Morgan's law) $= \overline{A} + 1$ $(B + \overline{B} = 1)$ the output always 1 irrespective of the logic which is an incorrect answer.

Third Boolean expression is $((A+B) + (A+B))' = (A+B)' = \overline{A+B}$ which is an incorrect answer.

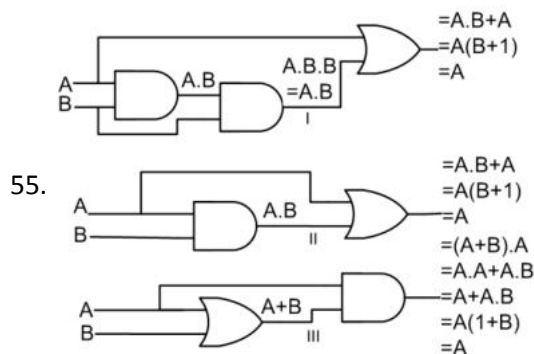
Fourth ones Boolean expression is $(A+B)' = \overline{A+B}$ as per De Morgan's law. For De Morgan's law may have a look at https://en.wikipedia.org/wiki/De_Morgan%27s_laws

Students can draw a truth table and comes to the conclusion that this refers to NAND gate hence they can choose the first answer.

A	B	$\overline{A}$	$\overline{B}$	$\overline{A+B}$
0	0	1	1	1
0	1	1	0	1
1	0	0	1	1
1	1	0	0	0



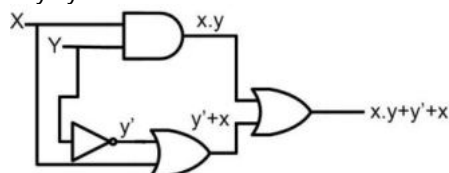
OR gates Boolean expression + for OR gates and . for AND gates. Substituting we will $X = (A+B).C.D + \overline{E}$



AL ICT students are able to solve this with the Boolean algebra. OL ICT students may draw the truth table each output and comes to the conclusion that all the circuits are the same.

56. A, B (Boolean expression is $= A \cdot \overline{X} + B \cdot X$ substituting 0, 1 respectively gives A, B alternatively substitute the values in the logic diagram).

57. $x.y + y' + x$



58. OL students can verify with the truth table

AL students can verify it with De Morgans Law $(A+b')' = A' \cdot B$ first answer.

59. $Q = A \cdot B$ Since $B=1$ So $Q = A$

60. NAND